

GS 21/L Physical and Structural Geology

Module 5 **Minerals and Rock Cycle**

COURSE OUTCOMES:

- At the end of this module, students will be able to:
- ✚ Analyze different surface geological features using basic physical geology principles.
- ✚ Classify different rock forming minerals and rock types typically found above and below the Earth's surface.

INTENDED LEARNING OUTCOMES:

- At the end of this module, the students will be able to:
- ✚ Identify different rock types according to composition, and environment of formation.
- ✚ Correlate the different environment of formation and the rock forming minerals with the three different rock types.

PROGRAM OUTCOMES:

- ✚ Apply knowledge of mathematics and science to solve complex petroleum engineering problems;
- ✚ design and conduct experiments, as well as to analyze and interpret data;
- ✚ identify, formulate, and solve complex petroleum engineering problems;

TIME FRAME:

- ✚ This module can be covered in a week.

MINERALS AND ROCK CYCLE

Mineral as any naturally occurring inorganic solid that possesses an orderly crystalline structure and a definite chemical composition that allows for some variation.

Mineral or not?

Ice

Coal

Petrified wood

Quartz

Sulfur

Gold

Coal

FIGURE 5.1 Quartz crystals

A collection of well developed quartz crystals found near Hot Springs, Arkansas. (Photo by Jeff Scovil)



MINERALS AND ROCK CYCLE

Characterized by:

1. **Naturally occurring** - minerals form by natural geologic processes.
2. **Inorganic** - inorganic crystalline solids, such as ordinary table salt (halite), that are found naturally in the ground are considered minerals.
3. **Solid substance** - only solid crystalline substances are considered minerals.
4. **Ordered internal structure** - Minerals are crystalline substances, which means their atoms are arranged in an orderly, repetitive manner.
5. **Definite chemical composition** – occurs because certain elements can substitute for others of similar size without changing the mineral's internal structure.

MINERALS AND ROCK CYCLE

ROCK - any solid mass of mineral, or mineral-like, matter that occurs naturally as part of our planet.

- some rocks are composed almost entirely of one mineral.
- some rocks are composed of nonmineral matter.



FIGURE 5.2 Most rocks are aggregates of minerals

Shown here is a hand sample of the igneous rock granite and three of its major constituent minerals.

(Photos by E. J. Tarbuck)

MINERALS AND ROCK CYCLE

How Do Minerals Form?

1. Precipitation of Mineral Matter

- most familiar way in which minerals grow is from an aqueous (water) solution containing dissolved material matter (ions). As long as the solution is not saturated, the motion of the dissolved ions keeps them from joining together.

- Two factors—a drop in temperature and water lost through evaporation— cause the solution to become closer to reaching saturation. Once saturation is reached, the ions begin to bond, forming solid crystalline solids (called salts) that precipitate from (settle out of) the solution.

MINERALS AND ROCK CYCLE

How Do Minerals Form?

2. Crystallization of Molten Rock

- crystallization of minerals from molten rock, although more complicated, is similar to water freezing.

- Crystallization of a molten mass generates igneous rocks that consist of a mosaic of intergrown crystals that tend to lack well developed faces.



FIGURE 5.3 Minerals form when molten rock solidifies

(Photo by Martin Rietze/ AGE Fotostock)

MINERALS AND ROCK CYCLE

How Do Minerals Form?

3. Deposition as a Result of Biological Processes

- Water-dwelling organisms are also responsible for transforming substantial quantities of dissolved material into mineral matter.

- Although most water-dwelling organisms produce hard parts made of calcium carbonate, some, such as diatoms and radiolarians, produce glasslike silica skeletons. During burial, this material forms microscopic silicon dioxide (quartz) crystals that are the main constituents of rocks such as chert and flint.



FIGURE 5.4 Minerals formed by biological processes

These limestone rocks formed in an ancient sea mainly from the remains of invertebrate animals that secreted shells composed of the carbonate minerals calcite and aragonite. (Andrew Komylak/Aurora RF/AGE Fotostock)

MINERALS AND ROCK CYCLE

Physical Properties of Minerals

1. *Optical Properties*

- a. **Luster** - the appearance or quality of light reflected from the surface of a mineral. Minerals that have the appearance of metals, regardless of color, are said to have a metallic luster. Most minerals have a nonmetallic luster and are described using various adjectives such as vitreous or glassy. Other nonmetallic minerals are described as having a dull, or earthy, luster or a pearly luster. Still others exhibit a silky luster or a greasy luster.
- b. **Ability to transmit Light** - When no light is transmitted, the mineral is described as opaque; when light, but not an image, is transmitted through a mineral sample, the mineral is said to be translucent. When both light and an image are visible through the sample, the mineral is described as transparent. (TRANSPARENT, TRANSLUCENT, OPAQUE)
- c. **Color** - it is considered a diagnostic property of only a few minerals. The use of color as a means of identification is often ambiguous or even misleading.
- d. **Streak** - the color of a mineral in powdered form. A mineral's streak is obtained by rubbing it across a streak plate (a piece of unglazed porcelain) and observing the color of the mark it leaves. Although the color of a mineral may vary from sample to sample, its streak is usually consistent in color.

MINERALS AND ROCK CYCLE

Optical Properties

The freshly broken sample of galena (right) displays a metallic luster, while the sample on the left is tarnished and has a submetallic luster.



FIGURE 5.5a
Metallic vs Submetallic luster
(Photo courtesy of E. J. Tarbuck)

Although the color of a mineral is not always helpful in identification, the streak, which is the color of the powdered mineral, can be very useful.

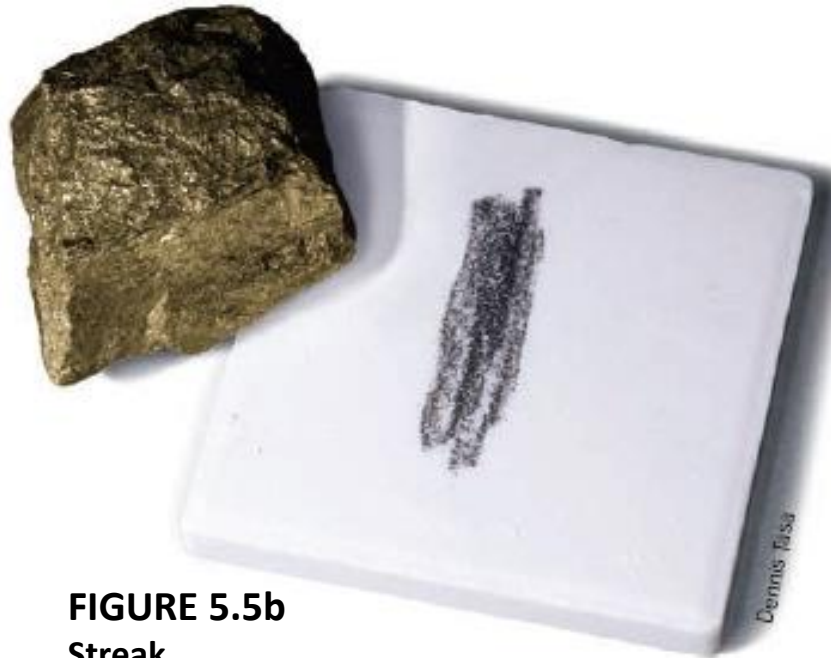


FIGURE 5.5b
Streak



A. Fluorite



B. Quartz

FIGURE 5.5c
Color variations in minerals
Some minerals, such as fluorite and quartz, occur in a variety of colors.
(Photo A by Dennis Tasa; photo B by E. J. Tarbuck)

MINERALS AND ROCK CYCLE

Physical Properties of Minerals

2. *Crystal Shape, or Habit*

- refers to the common or characteristic shape of a crystal or aggregate of crystals.

Examples:

Equant

Bladed

Fibrous

Tabular

Prismatic

Platy

Blocky

FIGURE 5.6

Some common crystal habits

A. Thin, rounded crystals that break into fibers.

B. Elongated crystals that are flattened in one direction.

C. Minerals that have stripes or bands of different color or texture.

D. Groups of crystals that are shaped like cubes.



A. Fibrous



B. Bladed



C. Banded



D. Cubic crystals

MINERALS AND ROCK CYCLE

Physical Properties of Minerals

3. *Mineral Strength*

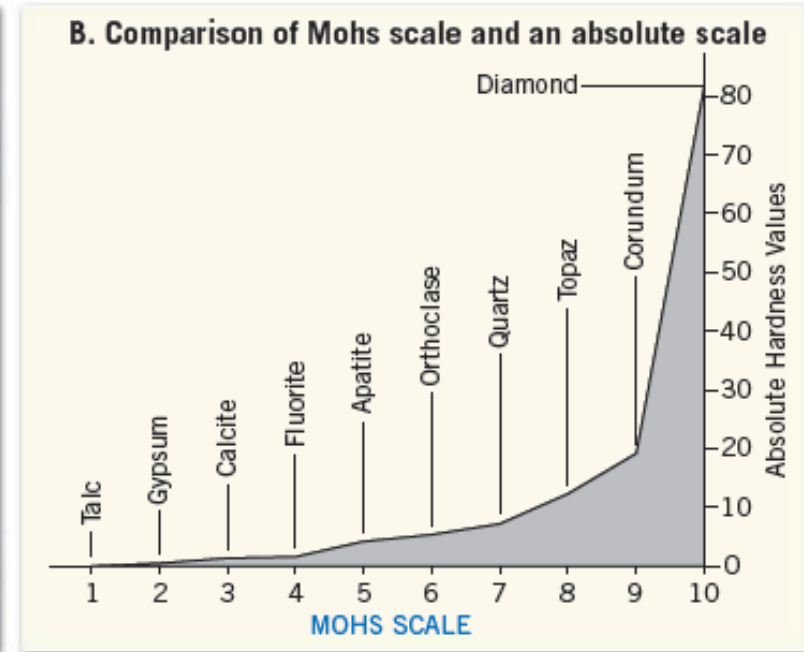
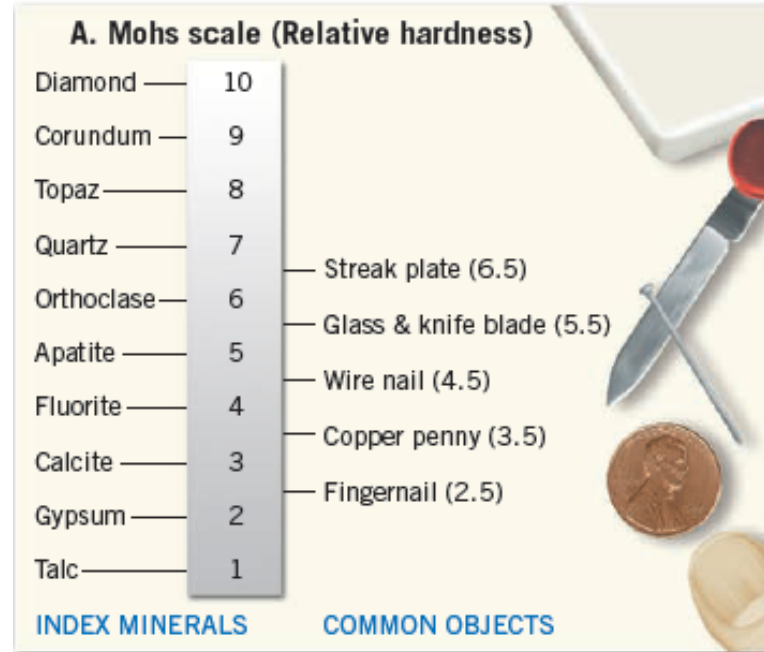
- how easily minerals break or deform under stress.

- a. **Tenacity** - describes a mineral's toughness, or its resistance to breaking or deforming.
- b. **Hardness** - a measure of the resistance of a mineral to abrasion or scratching, one of the most useful diagnostic properties.
- c. **Cleavage** - the tendency of a mineral to break (cleave) along planes of weak bonding.
- d. **Fracture** - minerals that have chemical bonds that are equally, or nearly equally, strong in all directions.
 - *IRREGULAR FRACTURE* (produce uneven surfaces)
 - *CONCHOIDAL FRACTURE* (break into smooth, curved surfaces)
 - *SPLINTERY AND FIBROUS FRACTURE* (produce splinters or fibers)

MINERALS AND ROCK CYCLE

Mineral Strength

FIGURE 5.7
Hardness scales



A. Irregular fracture



B. Conchoidal fracture

FIGURE 5.8
Irregular versus conchoidal fracture
(Photos by E. J. Tarbuck)

MINERALS AND ROCK CYCLE

Physical Properties of Minerals

4. *Density and Specific Gravity*

Density - mass per unit volume.

Specific gravity - dimensionless number representing the ratio of a mineral's weight to the weight of an equal volume of water. Most common minerals have a specific gravity between 2 and 3.

MINERALS EXAMPLES:

QUARTZ (SiO_2)- 2.5

METALLIC MINERALS I.E COPPER, MAGNETITE, PYRITE = more than twice the specific gravity of quartz

GALENA (PbS)- 7.5

MINERALS AND ROCK CYCLE

Other Properties of Minerals

MAGNETISM

DOUBLE REFRACTION - for example, when a transparent piece of calcite is placed over printed text, the letters appear twice.

EFFERVESCENCE

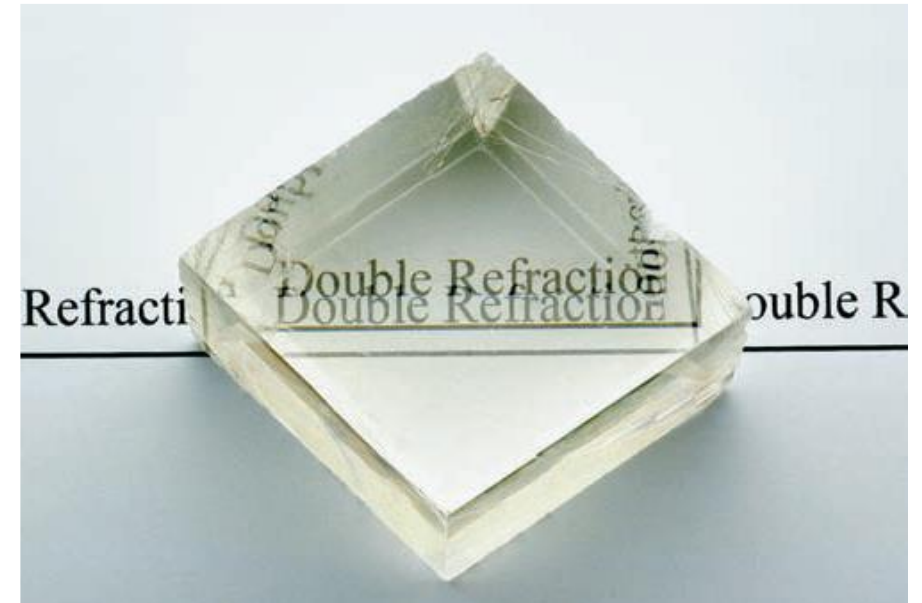


FIGURE 5.9 Double refraction

This sample of calcite exhibits double refraction. (Photo by Chip Clark)



FIGURE 5.10

Calcite reacting with a weak acid

(Photo by Chip Clark)

MINERALS AND ROCK CYCLE

Rock - forming Minerals

- Only eight elements make up the vast majority of the rock-forming minerals and represent more than 98 percent (by weight) of the continental crust.

COMMON ROCK FORMING MINERALS

1. **SILICATES**- Si and O are the major components
2. **NON SILICATES**- other mineral groups without Si and O.

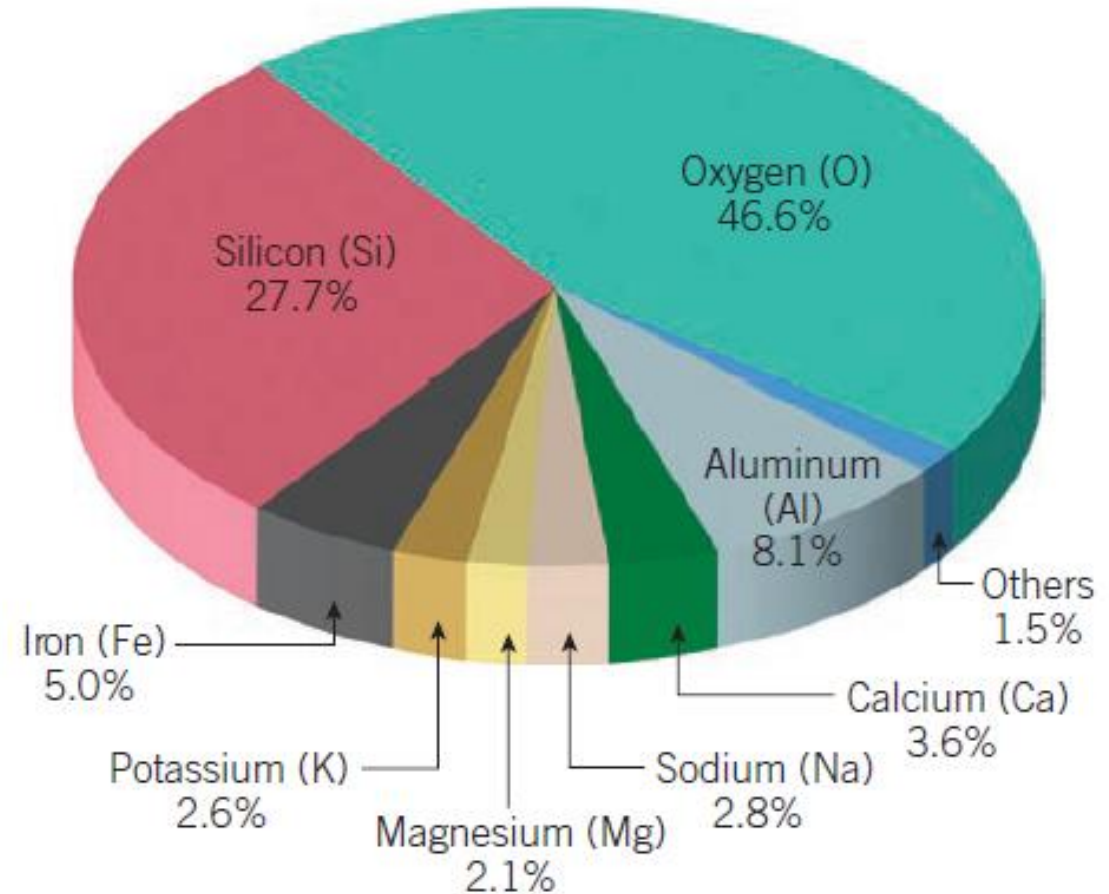


FIGURE 5.11

The eight most abundant elements in the continental crust

MINERALS AND ROCK CYCLE

The Silicates

- silicon–oxygen tetrahedron (SiO_4^{4-})
- This structure consists of four oxygen ions that are covalently bonded to one comparatively small silicon ion, forming a ***tetrahedron***—a pyramid shape with four identical faces.

Five Basic Silica Structures

1. Independent tetrahedra- *ex. Olivine group*
2. Single chain- *ex. Pyroxene group*
3. Double chain- *ex. Amphibole group*
4. Sheet structure- *ex. Mica group*
5. 3d network- *ex. Plagioclase group, quartz*

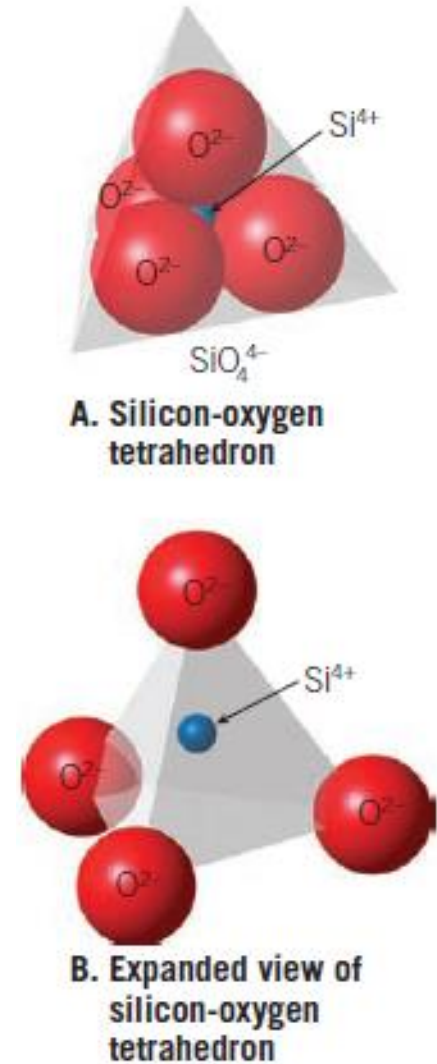
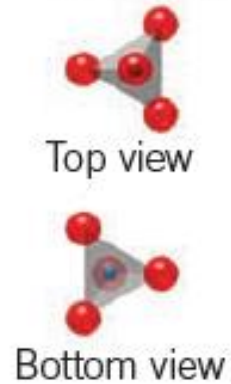


FIGURE 5.12

Two representations of the silicon–oxygen tetrahedron

MINERALS AND ROCK CYCLE

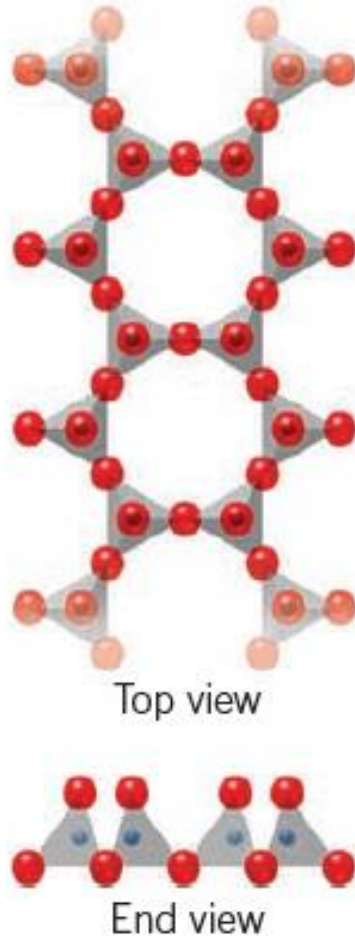
A. Independent tetrahedra



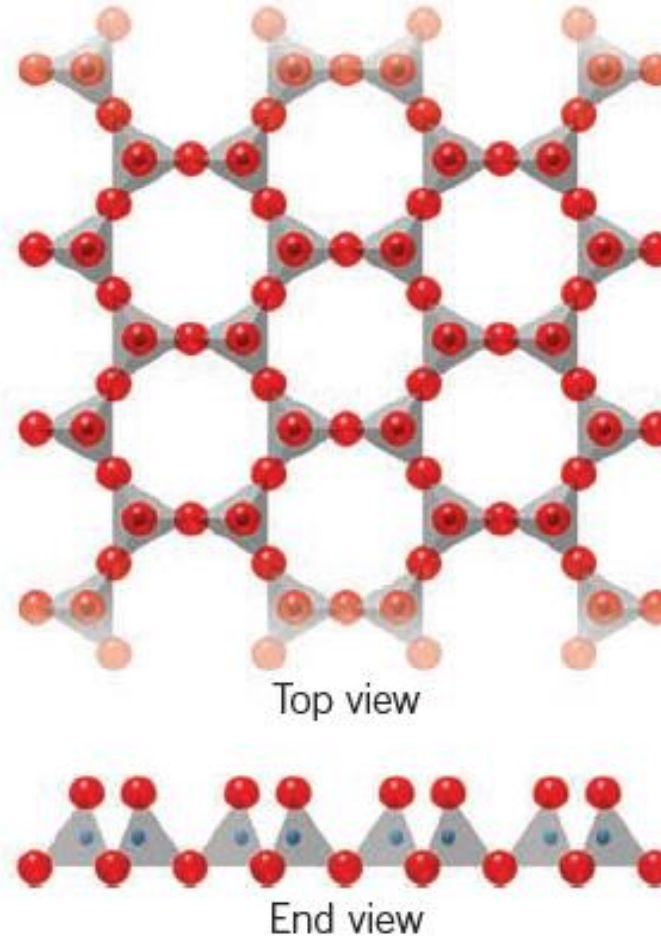
B. Single chain



C. Double chain



D. Sheet structure



E. Three-dimensional framework

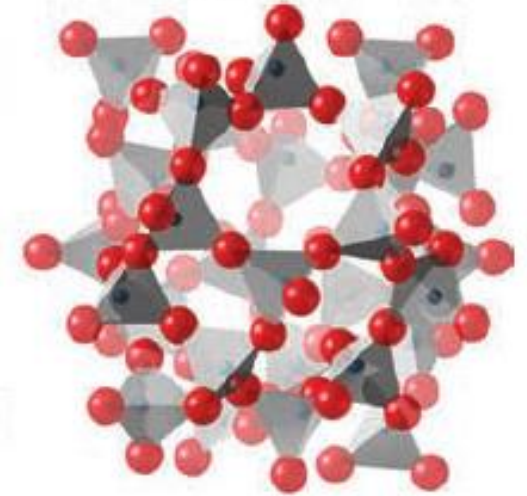


FIGURE 5.13
Five basic silicate structures

Common Silicate Minerals

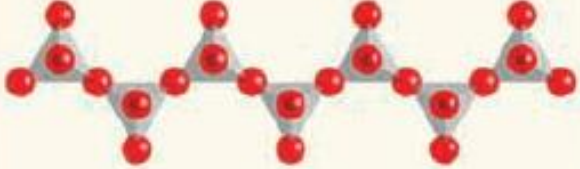
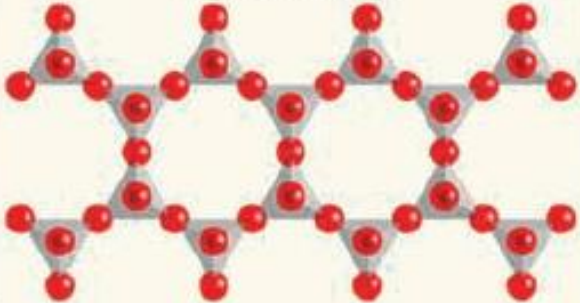
Mineral/Formula	Cleavage	Silicate Structure	Example
Olivine group $(\text{Mg, Fe})_2\text{SiO}_4$	None	Single tetrahedra 	 Olivine
Pyroxene group (Augite) $(\text{Mg, Fe})\text{SiO}_3$	Two planes at 90°	Single chains 	 Augite
Amphibole group (Hornblende) $\text{Ca}_2(\text{Fe, Mg})_5\text{Si}_8\text{O}_{22}(\text{OH})_2$	Two planes at 60° and 120°	Double chains 	 Hornblende

FIGURE 5.14

Common silicate minerals

Note that the complexity of the silicate structure increases from top to bottom.

(Photos by a: Marcel Clemens / Shutterstock; b: E. J. Tarbuck; c-g: Dennis Tasa)

Common Silicate Minerals

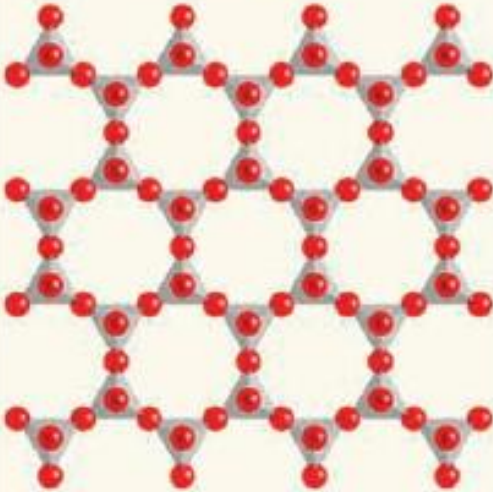
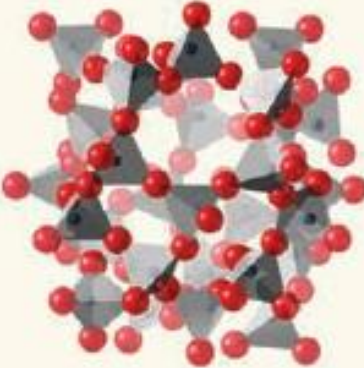
Mineral/Formula		Cleavage	Silicate Structure	Example
Micas	Biotite $K(\text{Mg, Fe})_3\text{AlSi}_3\text{O}_{10}(\text{OH})_2$	One plane	Sheets 	 Biotite
	Muscovite $\text{KAl}_2(\text{AlSi}_3\text{O}_{10})(\text{OH})_2$			 Muscovite
Feldspars	Potassium feldspar (Orthoclase) KAlSi_3O_8	Two planes at 90°	Three-dimensional networks 	 Potassium feldspar
	Plagioclase $(\text{Ca, Na})\text{AlSi}_3\text{O}_8$			
	Quartz SiO_2	None		 Quartz

FIGURE 5.14

Common silicate minerals

Note that the complexity of the silicate structure increases from top to bottom.

(Photos by a: Marcel Clemens / Shutterstock;
b: E. J. Tarbuck; c-g: Dennis Tasa)

Nonsilicate Minerals

Mineral Groups [key ion(s) or element(s)]	Mineral Name	Chemical Formula	Economic Use
Carbonates (CO ₃ ²⁻)	Calcite	CaCO ₃	Portland cement, lime
	Dolomite	CaMg(CO ₃) ₂	Portland cement, lime
Halides (Cl ¹⁻ , F ¹⁻ , Br ¹⁻)	Halite	NaCl	Common salt
	Fluorite	CaF ₂	Used in steelmaking
	Sylvite	KCl	Fertilizer
Oxides (O ²⁻)	Hematite	Fe ₂ O ₃	Ore of iron, pigment
	Magnetite	Fe ₃ O ₄	Ore of iron
	Corundum	Al ₂ O ₃	Gemstone, abrasive
	Ice	H ₂ O	Solid form of water
Sulfides (S ²⁻)	Galena	PbS	Ore of lead
	Sphalerite	ZnS	Ore of zinc
	Pyrite	FeS ₂	Sulfuric acid production
	Chalcopyrite	CuFeS ₂	Ore of copper
	Cinnabar	HgS	Ore of mercury
Sulfates (SO ₄ ²⁻)	Gypsum	CaSO ₄ · 2H ₂ O	Plaster
	Anhydrite	CaSO ₄	Plaster
	Barite	BaSO ₄	Drilling mud
Native elements (single elements)	Gold	Au	Trade, jewelry
	Copper	Cu	Electrical conductor
	Diamond	C	Gemstone, abrasive
	Sulfur	S	Sulfa drugs, chemicals
	Graphite	C	Pencil lead, dry lubricant
	Silver	Ag	Jewelry, photography
	Platinum	Pt	Catalyst

TABLE 5.1
Common
nonsilicate mineral
groups

Nonsilicate Minerals



A. Calcite



B. Dolomite



C. Halite



D. Gypsum



E. Hematite



F. Magnetite



G. Galena



H. Chalcopyrite



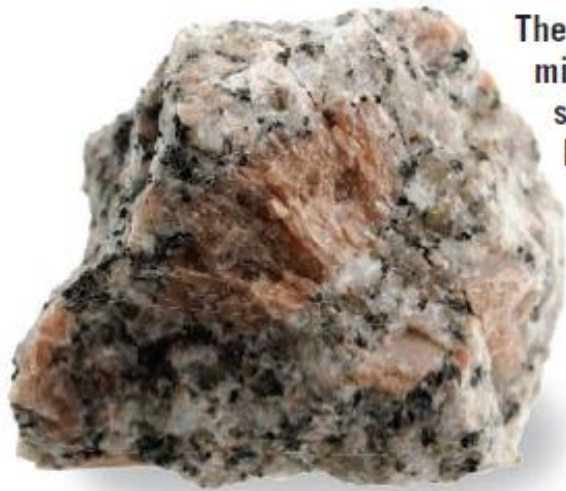
I. Fluorite

FIGURE 5.15
Important nonsilicate
minerals (Photos by Dennis
Tasa and E. J. Tarbuck)

MINERALS AND ROCK CYCLE

Rocks and the Rock Cycle

- The nature and appearance of a rock is strongly influenced by the minerals that compose it. In addition, a rock's texture—the size, shape, and/or arrangement of its constituent minerals—also has a significant effect on its appearance. A rock's mineral composition and texture, in turn, are a reflection of the geologic processes that created it.



The large crystals of light-colored minerals in granite result from the slow cooling of molten rock deep beneath the surface. Granite is abundant in the continental crust.



Basalt is rich in dark minerals. Rapid cooling of molten rock at Earth's surface is responsible for the rock's microscopically small crystals. Oceanic crust is composed mainly of basalt.

FIGURE 5.16

Two basic rock characteristics

Texture and mineral composition are basic rock features. Both rock samples are about golf-ball size (Basalt photo by Tyler Boyes/Shutterstock; granite photo by geoz/Alamy Images)

MINERALS AND ROCK CYCLE

Three Types of Rock



Igneous rocks form when molten rock solidifies at the surface (extrusive) or beneath the surface (intrusive). The lava flow in the foreground is the fine-grained rock basalt and came from SP Crater in northern Arizona.

FIGURE 5.17

Three rock groups

Geologists divide rocks into three groups - igneous, sedimentary, and metamorphic.

Sedimentary rocks consist of particles derived from the weathering of other rocks. This layer consists of durable sand-size grains of the glassy mineral quartz that are cemented into a solid rock. The grains were once a part of extensive dunes. This rock layer, called the Navajo Sandstone, is prominent in southern Utah.



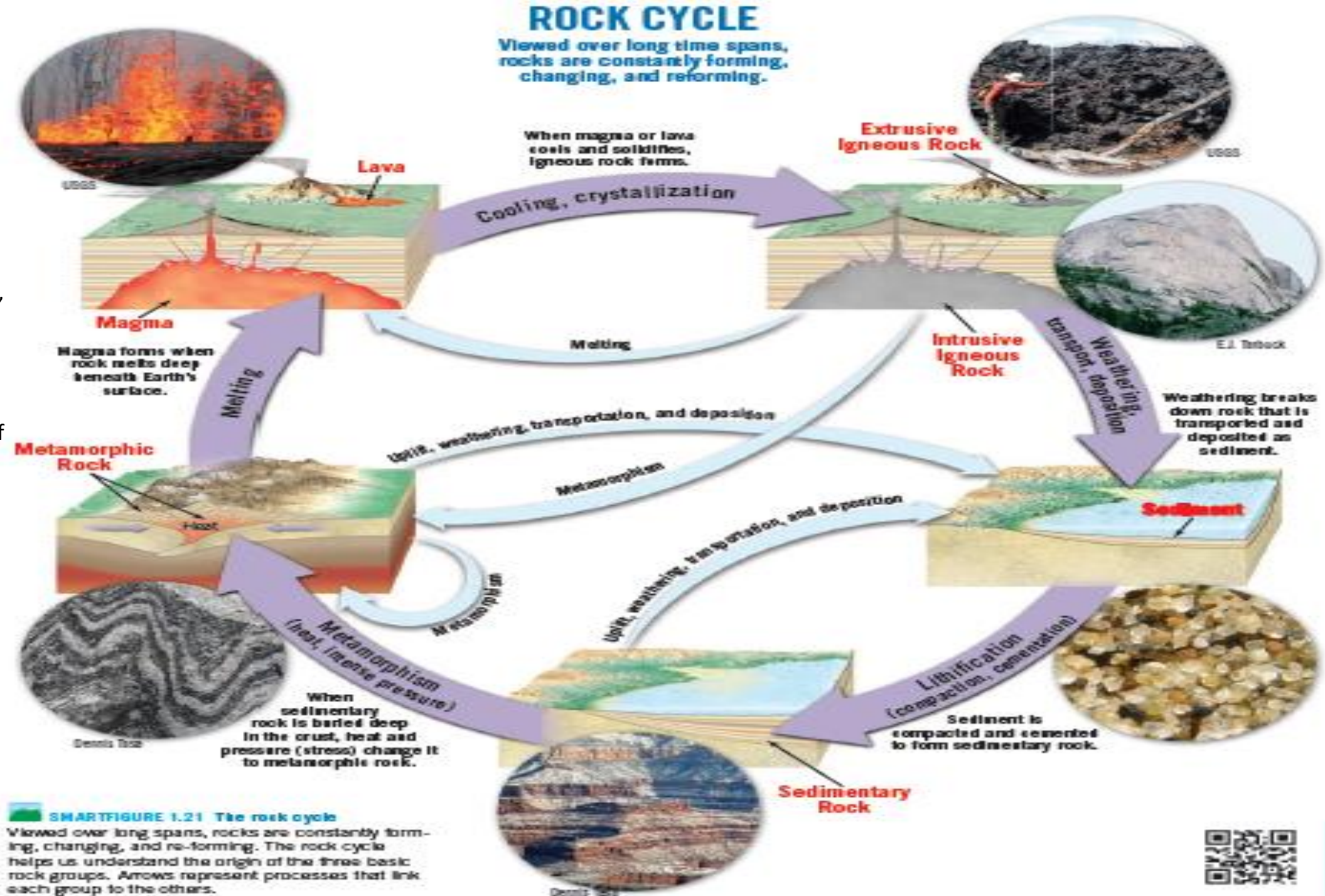
Dennis Tasa

FIGURE 5.18

The rock cycle

Viewed over long spans, rocks are constantly forming, changing, and re-forming. The rock cycle helps us understand the origin of the three basic rock groups. Arrows represent processes that link each group to the others.

https://media.pearsoncmg.com/bc/bc_0media_geo/smartfigure/sf-RockCycle.html



- Reference:
Tarbuck, Lutgens and Tasa (2014), Earth: An Introduction to Physical Geology 11th ed., Pearson Education, Inc